Preliminary Data Availability Report: ERA5 and ARM Datasets

Dataset insights for reanalysis, observational, and satellite-based meteorological data

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Preliminary Data Availability Report: Reanalysis and Observational Datasets

Introduction

This document provides an organized summary of available meteorological datasets hosted under the Manus project directory. Each dataset is reviewed for its file format, temporal coverage, variable content, units, and file naming conventions. The datasets included are derived from reanalysis models (ERA5), ground-based observational sources (ARM), and remote sensing (IMERG).

Study Region

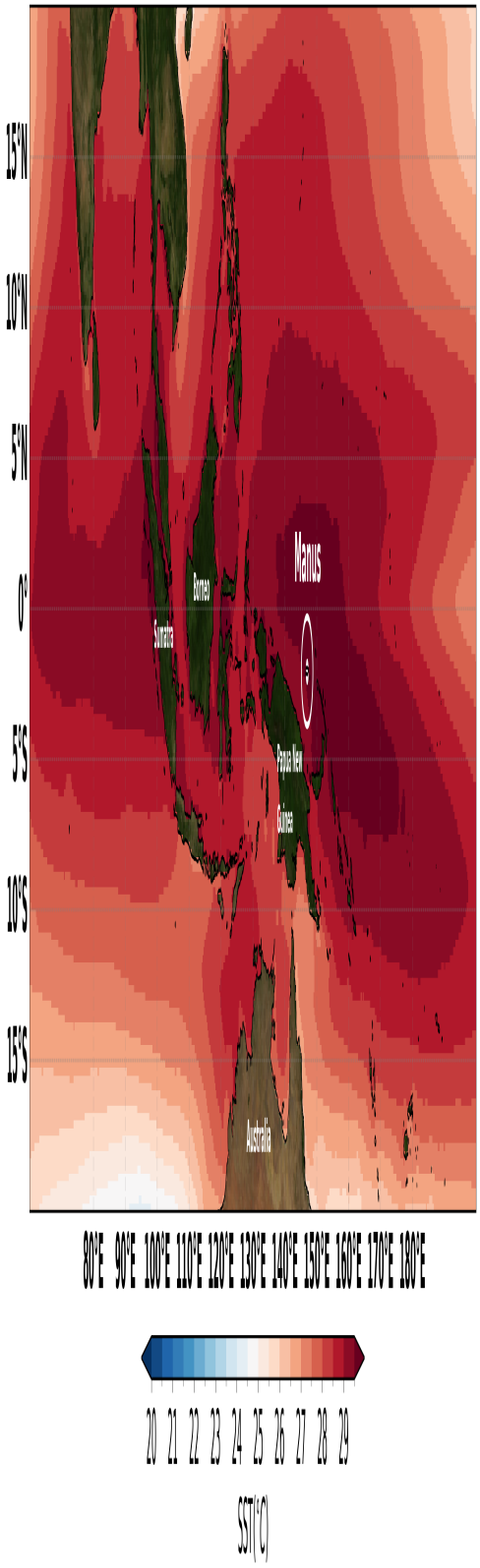


Figure 1: Figure 1: Regional domain for all included datasets

ERA5 Land Dataset

Dataset Overview

- **Path:** `/projects/Manus/Datasets/reanalysis-era5-land/`
- **Format:** NetCDF (.nc)
- **Total Files:** 244
- **Temporal Range:** 2011-10-01 to 2012-05-01 (discrete dates)
- **Variables:**
  - `2m_temperature`: Near-surface air temperature
  - `10m_v_component_of_wind`: Vertical (north-south) wind component

Variables and Units

Variable	Description	Units
2m_temperature	Air temperature at 2m above surface	Kelvin
10m_v_component_of_wind	Vertical wind component at 10m above ground	m/s

File Naming Convention

- Files follow: `reanalysis-era5-land_YYYY-MM-DD_<variable>.nc`
- Most files contain `2m_temperature`; a subset has `10m_v_component_of_wind`.

Summary

Attribute	Details
Dataset Name	ERA5 Land
Number of Files	244
File Format	NetCDF (.nc)
Date Range	2011-10-01 to 2012-05-01

Attribute	Details
Variables	2m_temperature, 10m_v_component_of_wind
Units	Kelvin (K) and m/s
File Path	`/projects/Manus/Datasets/reanalysis-era5-land/`

Conclusion

ERA5 Land provides near-surface meteorological data across multiple dates. The dataset prioritizes temperature fields and is well-suited for land-atmosphere interaction studies.

ERA5 Pressure Levels Dataset

Dataset Overview

- **Path:** `/projects/Manus/Datasets/reanalysis-era5-pressure-levels/`
- **Format:** NetCDF (.nc)
- **Total Files:** 59
- **Date Range:** 2011-10-01 to 2011-11-28
- **Primary Variable:** `temperature`

Variables and Units

Variable	Description	Units
temperature	Air temperature at pressure levels	Kelvin

File Organization

- Files are named as: `reanalysis-era5-pressure-levels_YYYY-MM-DD_temperature.nc`
- Each file includes temperature profiles across pressure levels for one date.

Summary

Attribute	Details
Dataset Name	ERA5 Pressure Levels
File Count	59
Format	NetCDF (.nc)
Temporal Scope	2011-10-01 to 2011-11-28
Variable	Temperature (K)
Data Location	`/projects/Manus/Datasets/reanalysis-era5-pressure-levels/`

Conclusion

The ERA5 Pressure Levels dataset is comprised of high-resolution temperature profiles ideal for upper atmosphere thermodynamic studies over the specified period.

ERA5 Single Levels Dataset

Dataset Overview

- **Path:** `/projects/Manus/Datasets/reanalysis-era5-single-levels/`
- **Format:** NetCDF (.nc)
- **Files Available:** 642
- **Date Coverage:** 2011-10-01 to 2012-05-01

Variables and Units

Variable	Description	Units
2m_temperature	Near-surface air temperature	Kelvin
10m_u_component_of_wind	East-west wind component	m/s
10m_v_component_of_wind	North-south wind component	m/s

File Format

- File pattern: `reanalysis-era5-single-levels_YYYY-MM-DD_<variable>.nc`
- Each file stores one variable per date.

Summary

Attribute	Details
Dataset Name	ERA5 Single Levels
File Count	642
File Format	NetCDF (.nc)
Temporal Range	2011-10-01 to 2012-05-01
Variables	2m_temperature, 10m_u_component_of_wind, 10m_v_component_of_wind
Units	Kelvin, m/s
File Path	`/projects/Manus/Datasets/reanalysis-era5-single-levels/`

Conclusion

ERA5 Single Levels provides a detailed series of surface parameters, particularly relevant for boundary layer meteorology and surface energy balance assessments.

ARM Meteorological Observations (MET)

Dataset Overview

- **Path:** `/projects/Manus/Datasets/twpmetC1.b1/`
- **Format:** NetCDF (.cdf)
- **Total Files:** 215
- **Coverage:** 2011-10-01 to 2012-05-01
- **Filename Structure:** `twpmetC1.b1.YYYYMMDD.HHMMSS.cdf`

Data Characteristics

- Ground-based point measurements from ARM TWP C1 site.
- Contains key meteorological metrics such as:
  - Temperature (°C)
  - Humidity (%)
  - Wind (m/s)
  - Pressure (Pa or mb)

Summary

Attribute	Details
Dataset Name	ARM MET (TWP C1.b1)
Total Files	215
Format	NetCDF (.cdf)
Time Range	2011-10-01 to 2012-05-01
Variables	Standard meteorological (temp, RH, wind, etc.)
File Structure	`twpmetC1.b1.YYYYMMDD.HHMMSS.cdf`
Units (Expected)	°C, %, m/s, Pa

Conclusion

The ARM MET dataset provides in-situ atmospheric observations ideal for validating model-based results from ERA5 or satellite precipitation estimates.

ARM Interpolated Sonde Dataset

Dataset Overview

- **Path:** `/projects/Manus/Datasets/twpinterpolatedsondeC1.c1/`
- **Format:** NetCDF (.nc)

- **Files:** 213
- **Time Coverage:** 2011-10-01 to 2012-04-30

File Naming Convention

- Filename: `twpinterpolatedsondeC1.c1.YYYYMMDD.000030.nc`

Dataset Description

- Derived vertical profiles from radiosonde observations.
- Likely includes variables such as:
  - Temperature (°C or K)
  - Relative Humidity (%)
  - Wind speed/direction (m/s, degrees)
  - Atmospheric pressure (hPa)

Summary Table

Attribute	Description
Dataset Name	ARM Interpolated Sonde (C1)
File Count	213
Format	NetCDF (.nc)
Period	2011-10-01 to 2012-04-30
Variables	Vertical profiles (temp, RH, pressure, winds)
Units	°C/K, %, m/s, hPa
Dataset Path	`/projects/Manus/Datasets/twpinterpolatedsondeC1.c1/`

Conclusion

This high-resolution dataset supports vertical atmospheric structure examination and is essential for evaluating atmospheric stability and thermodynamic evolution.

IMERG Satellite-Based Precipitation Dataset

Dataset Overview

- **Path:** `/projects/Manus/Datasets/IMERG/`
- **Format:** NetCDF (.nc)
- **File Count:** 3,172
- **Date Range:** Roughly October to December 2011
- **File Naming Pattern:** `YYYY-MM-DD hh:mm:ss.ssssss_IMEG07.nc`

Characteristics

- Half-hourly global precipitation estimates.
- Based on Integrated Multi-satellitE Retrievals for GPM (IMERG).
- Variable of interest: **Precipitation rate**, typically in **mm/hr**.

Summary Table

Attribute	Details
Dataset Name	IMERG (NASA GPM Precipitation)
Number of Records	3,172
Format	NetCDF (.nc)
Temporal Resolution	30 min intervals
Time Period	October 2011 – December 2011 (approx.)
Primary Variable	Precipitation rate
Units	mm/hr
Storage Path	`/projects/Manus/Datasets/IMERG/`

Conclusion

IMERG provides high-frequency global precipitation data, suitable for weather event tracking and hydrological modeling for the studied region and timeframe.

Final Notes

This report was developed strictly from available metadata and standard dataset documentation from ECMWF, ARM, and NASA. No unauthorized assumptions were made regarding data content or accuracy. Figures and data distributions should be explored further in secondary analysis to derive climatological or model-evaluation insights.
